

Remote Sensing Object Tracking With Deep Reinforcement Learning Under Occlusion

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Abstract—Object tracking is an important research direction of space Earth observation in the field of remote sensing. Although the existing correlation filter-based and deep learning (DL)-based object tracking algorithms have achieved great success, they are still unsatisfactory for the problem of object occlusion. The occlusion caused by the complex change in background, and the deviation of the tracking lens, causes object information to go missing, which leads to the omission of detection. Traditionally, most methods for object tracking under occlusion adopt a complex network model, which redetects the occluded object. To address this issue, we propose a novel object tracking approach. First, an action decision-occlusion handling network (AD-OHNet) based on deep reinforcement learning (DRL) is built to achieve low computational complexity for object tracking under occlusion. Second, the temporal and spatial context, the object appearance model, and the motion vector are adopted to provide the occlusion information, which drives actions in reinforcement learning under complete occlusion and contributes to improving the accuracy of tracking while maintaining speed. Finally, the proposed AD-OHNet is evaluated on three remote sensing video datasets of Bogota, Hong Kong, and San Diego taken from Jilin-1 commercial remote sensing satellites. The video datasets all shared problems of low spatial resolution, background clutter, and small objects. Experimental results on the three video datasets validate the effectiveness and efficiency of the proposed tracker.

Index Terms—Deep reinforcement learning (DRL), object tracking, occlusion, remote sensing datasets.

I. INTRODUCTION

OBJECT tracking involves calculating the exact position of an object in the next frame of a video, given the

object's coordinate position in the first frame. It is an important research direction of space Earth observation in the field of remote sensing and has broad applications in military monitoring, battlefield analysis, weapon navigation, missile warning systems, traffic management, and public security. In recent years, existing object tracking algorithms have achieved great success. However, due to the diversity of the characteristics of the tracked object and the complexity of the external environment, object tracking has always been a challenging task. A robust object tracking algorithm must be able to solve various difficulties encountered in the tracking process, such as appearance distortion, lighting changes, fast motion and motion blur, background clutter, out-of-plane rotation, in-plane rotation, scale variation, movement out of view, and occlusion.

It is generally accepted that conventional object tracking methods are divided into generative model methods and discriminative model methods. Currently, the most used method is the second one, which is generally known as the tracking-by-detection method. The generative method models the object area in the current frame and finds the most similar area in the next frame. Some well-known examples of such methods are the Kalman filter [1], particle filter [2], and mean shift [3]. For example, if it is known from the current frame that 60% of the object area is red and 40% is green, then the search algorithm blindly looks for an area that matches this color ratio in the next frame. However, for the discriminative model method, the current frame uses the object region as the positive sample and the background region as the negative sample. Then, machine learning is adopted to train the classifier, whose aim is to find the optimal region in the subsequent frame.

Currently, mainstream object tracking methods include correlation filter-based trackers and deep learning (DL)-based trackers, both of which belong to the discriminative model methods. The correlation filter-based trackers train the filters by regressing the input characteristics to the object's Gaussian distribution. In the follow-up trace, the response peak in the prediction distribution is searched to locate the position of the object. Minimum output sum of squared error (MOSSE) [4], circulant structure with kernels (CSK) [5], kernelized correlation filter (KCF) [6], and discriminative scale-space tracker (DSST) [7] are typical examples of such methods. Although these methods using low-level handcrafted features have gained attention due to their computational efficiency, they are vulnerable in dynamic situations (e.g., illumination changes, deformations, and occlusion) [8]. In recent years, an increasing number of researchers have focused on DL-based

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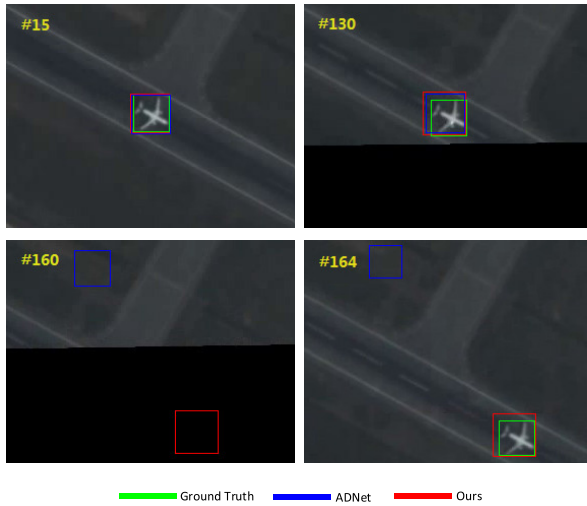


Fig. 1. Qualitative comparison of our tracker with the baseline ADNet. The number in the top-left corner represents the frame number within the video. Our AD-OHNet approach successfully tackles the issue of complete occlusion, leading to higher tracking precision.

trackers, such as the multidomain network [8], continuous convolution operator tracker (C-COT) [9], and efficient convolution operators [10]. Although such methods can automatically extract features through a deep network, the performance of tracking algorithms is still not satisfactory when encountering challenging object tracking tasks.

Solving the occlusion problem is one of the most important and challenging research directions in object tracking. As shown in Fig. 1, there may be deviations during mobile shooting on remote sensing platforms (such as satellites), resulting in partial or even complete occlusion of the photographed object. In recent years, domestic and overseas scholars have conducted research on object occlusion [11], [12]. However, there are still some problems to be solved. First, previous works have focused on videos of natural images rather than videos of remote sensing images. Compared with the natural image, the proportion of the object is smaller in the remote sensing image, and there is more background clutter and lower spatial resolution; therefore, it is more difficult to extract features from remote sensing images. Second, most tracking methods adopt a complex network model to solve the problem of occlusion, which must redetect the occluded object, and this is quite time-consuming. Finally, when the object is largely or even completely occluded, the position of the tracked object will be inaccurate or lost.

To solve the aforementioned problems, we adopt the framework of deep reinforcement learning (DRL) to model remote sensing object tracking under complete occlusion. The algorithm combines the object occlusion information provided by the temporal and spatial context, the object appearance model, and the motion vector, which significantly increases the robustness of the tracking algorithm. In sum, the proposed method aims to address the problems of complete occlusion for single-object tracking. The main contributions are summarized as follows.

- 1) First, the framework of DRL is adopted to perform object tracking without an additional network structure

to improve the speed of the algorithm under occlusion. The state and action parameters in DRL can correlate with the information of continuous frames in object tracking. To the best of our knowledge, this is the first time that the idea of DRL has been used to address the occlusion problem in object tracking.

- 2) Second, we propose a novel approach that drives action by using the object occlusion information in a tracking task, which is unlike traditional tracking-by-detection methods. The temporal and spatial contexts between the consecutive frames in the remote sensing sequence, the object appearance model learned by the network, and the motion vector obtained from action parameters in reinforcement learning are adopted to provide the occlusion information, which remarkably enhances the robustness and precision of the object tracking algorithm.
- 3) Finally, the action decision-occlusion handling network (AD-OHNet) approach, proposed primarily for complete occlusion rather than partial occlusion, is evaluated on remote sensing video datasets instead of natural image video datasets. According to the characteristics of the direction of object movement in the remote sensing datasets, the type of action is changed. The results show that the proposed method achieves excellent performance.

The remainder of this article is organized as follows. Brief introductions of previous works related to the object tracking methods are presented in Section II. Section III provides the details of object tracking with DRL under occlusion. In Section IV, experimental results and analysis are presented. Finally, conclusions are drawn in Section V and are followed by a brief discussion of directions for future work.

II. RELATED WORK

Research on object tracking has achieved great advances in the past few decades, and scholars have developed various state-of-the-art tracking algorithms. In this section, we will review some related work for object tracking in terms of object tracking, occlusion, DRL in object tracking, and remote sensing object tracking.

A. Object Tracking

At present, the main tracking algorithms include correlation filter-based trackers and DL-based trackers. We can refer to the object tracking benchmark in [13] and [14] for a systematic review.

1) *Correlation Filter-Based Trackers*: The correlation filter is still a focus of object tracking research and originated in the field of signal processing. The fast Fourier transform was introduced into this method, which greatly improved its speed. The earliest correlation filter algorithm used for object detection was the MOSSE [4]. As the first algorithm used in the correlation filter, it laid the foundation for subsequent algorithms. CSK [5] was proposed based on MOSSE. The kernel-based tracking method using circulant matrices mathematically solved the problem of dense samples. However,

there are some problems with CSK, such as scale variation, and a marginal performance when using the input of grayscale images and circulant matrices. KCF and dual correlation filter (DCF) [6] algorithms were optimized on multichannel features and kernel methods. The input of CSK was a single-channel feature grayscale image, but the inputs of KCF and DCF were multichannel feature images and could contain a color feature or histogram of oriented gradients (HOG) feature. Color name (CN) [15] was proposed later. CN could extract multicolor channel features, which was a great improvement over the previous HOG. The method proposed by Danelljan *et al.* [16] learned discriminative correlation filters based on a scale pyramid representation for estimating translation and scale independently. Afterward, they proposed DSST [7], which was capable of handling multiscale object tracking. Subsequently, the spatially regularized discriminative correlation filters [17] algorithm overcame boundary effects and penalized correlation filter coefficients with spatial regularization. Most recently, Zhang *et al.* [18] introduced a multitask correlation particle filter for robust object tracking. DL is a hot topic currently, and DL-based trackers have demonstrated competitive performance against correlation filter-based trackers.

2) *DL-Based Trackers*: DL has a strong ability for feature extraction and learning required features by itself. Moreover, an end-to-end DL network framework can be established easily. There are some object tracking methods based on DL, and most are focused on discriminatively learning object appearance models. For example, Hong *et al.* [19] proposed an online object tracking algorithm by learning a discriminative saliency map using a convolutional neural network (CNN). Given a pretrained CNN offline, this method utilized features to learn object appearance models using an online support vector machine (SVM). Nam and Han [8] also pretrained a CNN to obtain a generic object representation. Their algorithm, referred to as the multidomain network, was comprised of shared layers, and multiple branches of domain-specific layers corresponded to individual training sequences in the network. At the same time, Nam *et al.* [20] presented an algorithm for object tracking by managing multiple object appearance models in a tree structure. Bertinetto *et al.* [21] combined two image patch representations to learn a model that is inherently robust to both color changes and deformations. Meanwhile, Lu *et al.* [22] proposed a cascade tracker to promote the robustness of the tracker and reduce time consumption. Other tracking methods based on DL are an extension of correlation filtering methods. For example, Danelljan *et al.* [9] introduced a formulation for learning a discriminative convolution operator in the continuous spatial domain. To resolve the problem of computational complexity and overfitting, Danelljan *et al.* [10] then proposed efficient convolution operators for tracking. Moreover, there has been a lot of research on using the Siamese network for object tracking. For instance, Bertinetto *et al.* [23] introduced fully convolutional Siamese networks for object tracking. Meanwhile, Valmadre *et al.* [24] utilized an asymmetric Siamese network to realize end-to-end representation learning for correlation filter-based tracking. Bertinetto *et al.* [23] equipped a basic tracking algorithm with a novel fully

convolutional Siamese network and focused on learning strong embeddings in an offline phase. Zhang *et al.* [25] proposed a structure-constrained part-based model for tracking using deep neural networks. The residual attentional Siamese network [26] was employed with different kinds of attention mechanisms for high-performance object tracking. In addition, Li *et al.* [27] proposed a gradient-guided network to exploit the discriminative information in gradients and updated the template in the Siamese network through feedforward and backward operations. Most recently, Chen *et al.* [28] exploited the expressive power of the fully convolutional network and proposed a visual tracking framework (named Siamese box adaptive network), which directly classifies objects and regresses bounding boxes in a unified network.

B. Occlusion

Occlusion is a common and challenging problem in object tracking. When the object is partially or completely occluded, the reduction in object feature information may lead to the failure of the tracking algorithm and the detection of false objects. A robust object tracking algorithm should be able to accurately determine the occurrence of occlusion and use the feature information of the object to continue tracking the object during the occlusion period. Whether it can effectively handle occlusion, especially severe occlusion and complete occlusion, is important for evaluating object tracking algorithms. In this study, we aim at the complete occlusion in object tracking resulting from deviation of tracking lens.

Over the past decades, some works have focused on the problem of occlusion in object tracking. Yilmaz *et al.* [11] proposed a Bayesian framework for contour tracking with occlusion handling at an early stage, in which object shape priors were used to recover the missing object regions during occlusion. Then, Yin and Collins [29] achieved tracking after occlusion via numerical hybrid local and global mode seeking and applied cluster analysis to the sampled parameter space to redetect the object and renew the local tracker. Khan and Gu [30] used a nonlinear dynamic model for partial occlusion object tracking with Bayesian online learning on Grassmann manifolds. Later, Zhao *et al.* [31] addressed the impact of occlusion from the background by estimating the background in complex conditions with moving cameras while retaining adaptability to stationary cameras. Zhang *et al.* [32] introduced a kernel multiframe multifeature joint modeling tracker to solve the problem of occlusions, illumination changes, and pose variations by exploiting multiple feature cues and temporal consistency in a unified framework. Guan *et al.* [33] proposed an event-triggered tracking framework to coordinate the correlation tracking module, object redetection module, and occlusion and drift identification module. More recently, Li *et al.* [34] proposed a long-term tracking strategy to deal with occlusion that utilizes a context-aware correlation filter-based tracker with an added SVM for redetection.

However, none of the aforementioned methods are based on DL to solve the problem of occlusion. Trackers using DL are not very mature, and the occlusion problem still needs

to be solved. There has been related research reported by several scholars. For example, Zhou *et al.* [35] proposed a deep alignment network based on a multiperson tracking method with occlusion and motion reasoning. The deep alignment network-based appearance model and a Kalman filter-based motion model were adopted to handle the occlusion. In order to robustly deal with complex tracking scenes, Wu *et al.* [36] combined an adaptive Kalman filter with a Siamese region proposal network to make full use of the object's spatial-temporal information. The hard example discrimination method was proposed to effectively judge the occlusion situation and improve the Kalman filtering mechanism for adaptively updating tracker model. Yuan *et al.* [37] utilized ResNet to extract more robust features. The response maps computed from the different layers of ResNet were weighted and fused using the AdaBoost algorithm for accurate localization during tracking. Moreover, an optimized update strategy was proposed to handle the problem of object occlusion. In these methods, a more complex network model or strategy was usually adopted to achieve redetection of the object under occlusion. However, in this article, we propose a remote sensing object tracking method based on DRL to deal with complete occlusion.

C. DRL in Object Tracking

As a machine learning technique, reinforcement learning uses the trial and error mechanism in human learning [38]. The technique emphasizes learning while interacting with the environment and uses feedback to optimize decision-making. Reinforcement learning needs to discover appropriate actions to take in a given situation to maximize the reward. DRL is one of the most researched topics in artificial intelligence [39]. It combines the perceptual ability of DL with the decision-making ability of reinforcement learning and directly controls the behavior of the agent through the learning of high-dimensional perceptual input. DRL is beneficial for solving the perceptual decision-making problems of complex systems. In the object tracking task, the tracker has to make decisions on the moving direction and distance of the object in each frame of the video.

Recently, several scholars have conducted research on DRL for object tracking. Supancic and Ramanan [40] formulated tracking as a partially observable decision-making process and used reinforcement learning to learn decision-making strategies in a data-driven way. Xie *et al.* [41] utilized DRL for the selection of an optimal correlation filter model among multiple correlation filter models in parallel. Yun *et al.* [42] proposed the action-driven deep tracker, which pursued the object by sequential actions iteratively, and made a major contribution to the reduction of computation complexity in tracking. Dong *et al.* [43] presented a hyperparameter optimization algorithm for object tracking using continuous deep Q-learning. Luo *et al.* [44] introduced an end-to-end active tracker via DRL for reducing human workload in image labeling or tuning trail-and-error systems in the real world. In the present study, we propose a method based on DRL that can handle the problem of complete occlusion for remote sensing object tracking.



Fig. 2. Two types of images. (a) Natural image from VOT2015. (b) Remote sensing image from Bogota.

D. Remote Sensing Object Tracking

Although many object tracking methods have been developed, the research task of object tracking is very challenging in remote sensing videos. Compared with natural images, remote sensing images are quite different, such as being of larger scale, containing a large field of view, containing small objects, having background clutter, and being of low spatial resolution. For clarity, a natural image and a remote sensing image are shown in Fig. 2. There are several difficulties for object tracking in remote sensing video. First, the shape of the object is complex. Due to different scales, arrangement angles, rapid changes in direction, light distributions, illumination changes, and the multitemporal nature of weather conditions when collecting image data, it is difficult to summarize the object's regularity. For example, most of the airplanes in Fig. 2(b) are tilted. Second, background clutter is more visible. More background clutter similar to image features of the object will appear in the high-resolution remote sensing image data because of the large imaging scale of the optical satellite. For example, Fig. 2(b) does not have as many different colors as Fig. 2(a). In addition, complex man-made buildings and parking apron, especially for textures and shadows, have caused great interference to track when the object is an airplane in an airport. Finally, remote sensing videos contain a large number of small objects, such as the airplanes in Fig. 2(b), which makes feature extraction very difficult. For example, a remote sensing image is 285×220 pixels, but the object's size is only 28×10 pixels in a remote sensing video.

Some research in this field has been recently conducted. For example, Bi *et al.* [45] developed a tracking method based on a saliency-enhanced multidomain CNN to improve tracking performance for remote sensing videos. In this article, an object tracking method based on DRL is proposed to deal with the problem of complete occlusion in remote sensing video.

III. AD-OHNET APPROACH

In this section, we introduce the network framework for object tracking on remote sensing video benchmarks. The architecture of the proposed AD-OHNet is shown in Fig. 3.

A. Problem Settings

The object tracking problem can be described as a Markov decision process (MDP), which contains a set of actions $\mathcal{A}$, a set of states $\mathcal{S}$, transition dynamics $\Gamma(s_{t+1}|s_t, a_t)$, and a

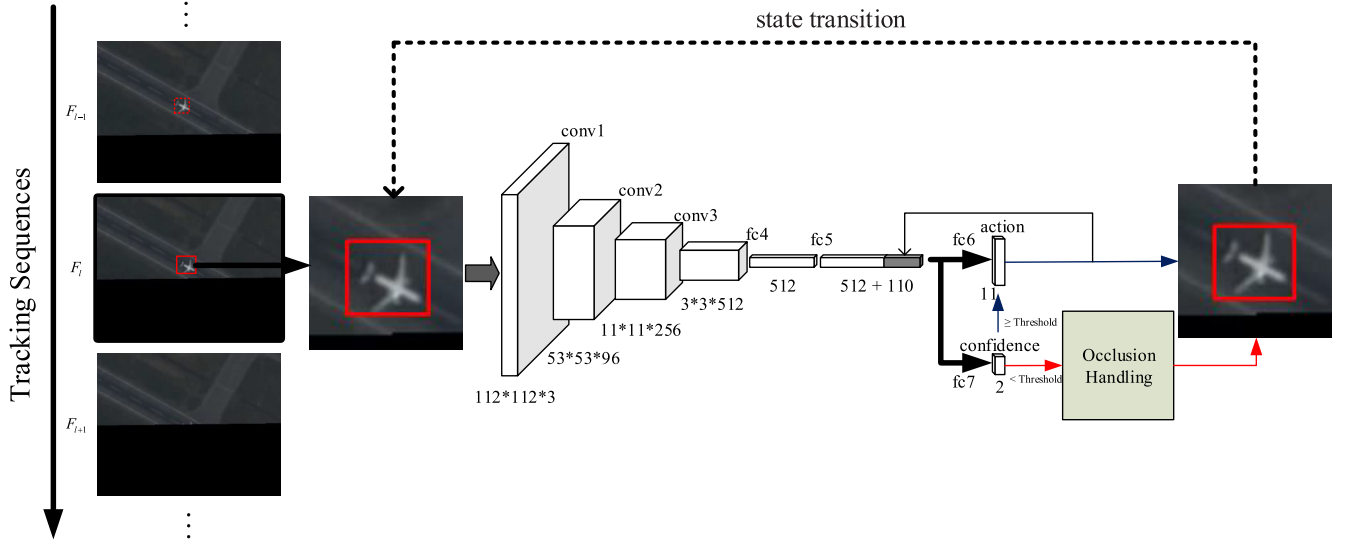


Fig. 3. Architecture of our proposed AD-OHNet. The dashed lines indicate the state transition. When the confidence score of the output by the fully connected layer fc7 is less than the threshold, the object is judged to be in the occlusion state, and the tracker enters occlusion handling. Otherwise, the tracker obtains the moving direction of the object bounding box in the current frame according to the action output from the network fully connected layer fc6. This process is iterated until the tracker has detected the object of all frames in the video sequence.

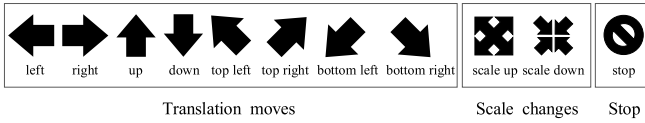


Fig. 4. All types of actions defined in our tracker.

reward function $\mathcal{R}(s_t, a_t, s_{t+1})$. The tracker is regarded as an agent, which aims to acquire an object with a bounding box. The tracker can obtain the state and action of each frame through iteration.

1) *Action*: There are three types of actions [42]: translation moves, scale changes, and stopping (see Fig. 4). The translation moves contain eight directions, namely, left, right, up, down, top left, top right, bottom left, and bottom right. The scale changes include scale up and scale down. Finally, stopping occurs when there is no more action, and tracker receives a reward in the current frame. Therefore, there are 11-D vectors for each action.

2) *State*: The state consists of two parts: the appearance model of the object and action dynamics. The appearance model of the object refers to the image patch within the bounding box, which is cropped from the original image in the video frame. There is some information of the coordinate position of the top-left corner, width, and height in bounding box expressed as $[x^{(t)}, y^{(t)}, w^{(t)}, h^{(t)}]$, where t ($t = 1, \dots, T$) denotes the t th iteration. The action dynamics concatenate previous k actions, where k is 10 in our experiments.

3) *Transition Dynamics*: After taking a decision action a_t in state s_t , the tracker acquires the next state s_{t+1} by transition dynamics. At the same time, the bounding box needs to be moved according to the corresponding action. The amount of movement can be expressed as

$$\Delta x^{(t)} = \alpha w^{(t)} \quad \text{and} \quad \Delta y^{(t)} = \alpha h^{(t)} \quad (1)$$

where $\alpha = 0.03$ in our experiments. For example, if the right action is selected, the position of the patch moves to $[x^{(t)} + \Delta x^{(t)}, y^{(t)}, w^{(t)}, h^{(t)}]$ from $[x^{(t)}, y^{(t)}, w^{(t)}, h^{(t)}]$. Moreover, the action dynamics are updated iteratively in each state transition.

4) *Reward*: The reward reflects whether the tracker successfully tracks the object. The reward is zero during iteration in MDP in a frame. The intersection over union (IoU) represents the overlap ratio between the tracked bounding box and the ground truth. The IoU is defined by

$$\text{IoU} = \frac{\text{BoundingBox} \cap \text{GroundTruth}}{\text{BoundingBox} \cup \text{GroundTruth}}. \quad (2)$$

When the stop action is selected, we define the reward function as

$$r(s_T) = \begin{cases} 1, & \text{if IoU} > 0.7 \\ -1, & \text{otherwise.} \end{cases} \quad (3)$$

Fig. 5 shows the qualitative results obtained with different IoUs. We can see that the bounding box obtained by the tracker can basically detect the object when the IoU is 0.7. The example in Fig. 5 is in the case of minimum $\text{BoundingBox} \cup \text{GroundTruth}$. In the actual object tracking, $\text{BoundingBox} \cup \text{GroundTruth}$ is larger, which means that $\text{BoundingBox} \cap \text{GroundTruth}$ is larger, and better tracking performance is obtained with the same value of IoU (see Fig. 6). Therefore, we assume that the tracker has fulfilled its task when $\text{IoU} > 0.7$, and the reward $r(s_T)$ is assigned to 1 if $\text{IoU} > 0.7$.

B. Tracking Algorithm

Refer again to the architecture of the proposed AD-OHNet shown in Fig. 3. The image patch is cropped from the current video frame to match the input size of our AD-OHNet.

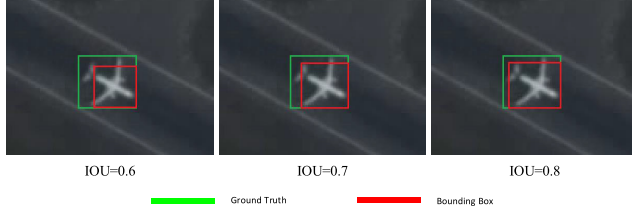


Fig. 5. Comparison of different IoUs.

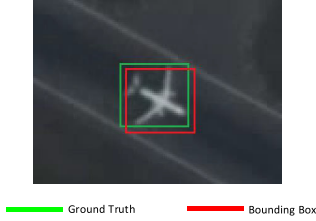


Fig. 6. Example when IoU = 0.7.

The network has three convolutional layers {conv1, conv2, conv3}, and two fully connected layers {fc4, fc5} combined with ReLU and dropout layers. The output of the fc5 layer is concatenated with the action dynamics vector. Finally, the fc6 layer predicts the probability of each action and selects the maximum value as the current action to be executed. The confidence scores are obtained through the fc7 layer to determine whether the object is tracked correctly. When the confidence score of the output by fully connected layer fc7 is less than the threshold, the object is judged to be in the occlusion state and the tracker enters the occlusion handling stage. Otherwise, the tracker obtains the moving direction of the object bounding box in the current frame according to the action output from the network fully connected layer fc6. This process is iterated until objects of all frames are detected in the video sequence.

The original weights of the network are derived from a pretrained network model in our work, which saves a lot of time and computing resources. Training samples are generated to train network parameters $\{w_1, w_2, w_3\}$ for the convolutional layer and $\{w_4, w_5, w_6, w_7\}$ for the fully connected layer. Training samples include image patches $\{p_i\}$, action labels $\{o_i^{(a)}\}$, and class labels $\{o_i^{(c)}\}$. The sample patch $\{p_i\}$ is obtained by adding Gaussian noise to the ground truth. The action label $\{o_i^{(a)}\}$ is assigned by

$$o_i^{(a)} = \arg \max_a \text{IoU}(f(p_i, a), G) \quad (4)$$

where G is the ground truth patch and $f(p_i, a)$ denotes the moved patch from $\{p_i\}$ by the action a . The class label $\{o_i^{(c)}\}$ is defined by

$$o_i^{(c)} = \begin{cases} 1, & \text{if IoU} > 0.7 \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

The network parameters are trained by minimizing the loss function by stochastic gradient descent. The loss function is defined by

$$\text{Loss} = \frac{1}{m} \sum_{i=1}^m L(o_i^{(a)}, \hat{o}_i^{(a)}) + \frac{1}{m} \sum_{i=1}^m L(o_i^{(c)}, \hat{o}_i^{(c)}) \quad (6)$$

where m denotes the batch size, L denotes the cross-entropy loss, and $\{\hat{o}_i^{(a)}\}$ and $\{\hat{o}_i^{(c)}\}$ denote the action and class predicted by the network, respectively. At the same time, the online fine-tuning strategy [8] is adopted to adapt to the shape change and deformation of the object in the test sequences, which makes the tracking algorithm more robust. w_i denotes the parameters of the i th layer, and W represents the whole network parameter. When the network is fine-tuned online, the parameters of the convolutional layer $\{w_1, w_2, w_3\}$ are fixed, and the parameters of the fully connected layer $\{w_4, w_5, w_6, w_7\}$ are fine-tuned. Whenever m -frames (here $m = 30$) have been detected or tracking fails, the tracker will enter the online fine-tuning process. During this process, the positive and negative samples will be updated.

Our task is to track a single object in a remote sensing video, given the location information of the object in the first frame. The position of the tracked patch is used to generate the training sample for the subsequent frames if the confidence value obtained by the fc7 layer is greater than 0.5. Each frame is detected n times in the remote sensing video unless the confidence score is less than 0.5 or the acquired action is the stopping action. n is obtained through many experiments. The number of actions per frame is about 15 when the object is not occluded in the video sequence. Here, we take the maximum value of 20 as the value of n in our experiments. Therefore, multiple actions can be performed in one frame, and the number of actions varies for different datasets. After each detection, an 11-D vector is obtained; then, the maximum value of the vector is selected, and the corresponding action is performed. This process is iterated continuously until the object tracking in the whole video is completed. However, when the confidence score is below 0.5, it can be concluded that the tracker has not tracked the object. In this article, this leads to occlusion handling, which is described in the following. Meanwhile, we summarize the complete procedure for object tracking in Algorithm 1.

C. Occlusion Handling

In this section, we introduce an approach to tackle the problem of complete occlusion in single-object tracking. Object occlusion has always been a common and difficult problem in tracking. Inspired by the temporal network and spatial network in [46], the temporal and spatial information is exploited to boost object tracking under occlusion in this article.

When the confidence score is less than 0.5 in Fig. 3, it is determined that the tracking has failed, and this means progressing to occlusion handling. The temporal and spatial context, object appearance model, and motion vector are adopted to drive actions of reinforcement learning.

1) *Temporal and Spatial Context*: The temporal and spatial information is provided for the current frame by the statistics of the actions and states before complete occlusion. The most important step is to calculate the average moving direction and distance of the object in each frame. If the bounding box at frame t is represented as $[x^{(t)}, y^{(t)}, w^{(t)}, h^{(t)}]$, the bounding box at frame $t + 1$ is represented as $[x^{(t+1)}, y^{(t+1)}, w^{(t+1)}, h^{(t+1)}]$, that is to say,

Algorithm 1 AD-OHNet Tracking Algorithm**Require:**

Pre-trained network filters, $\{w_1, w_2, w_3, w_4, w_5, w_6, w_7\}$;
Initial bounding box of the object, b_1 ;

Ensure:

Predicted bounding boxes of the object, $\mathcal{B}$;
1: Extract positive samples N_1^+ and negative samples N_1^- ;
2: Initial weight $\{w_4, w_5, w_6, w_7\}$ of the full connection layer using N_1^+ and N_1^- ;
3: **for** $t \in [1, T]$ **do**
4: Extract $112 * 112$ image patch, p_t , as the input of the network;
5: Obtain the actions $\mathcal{A}$ and confidence score $\mathcal{C}$ through the network;
6: Compare the size of confidence score $\mathcal{C}$ and threshold T ;
7: **if** $\mathcal{C} \geq T$ **then**
8: Update positive samples N_t^+ and negative samples N_t^- ;
9: Update weight $\{w_4, w_5, w_6, w_7\}$ of the full connection layer;
10: Obtain the moving direction and distance of the object according to actions $\mathcal{A}$;
11: Obtain the state $\mathcal{S}$ and bounding box b_t of the object;
12: **else**
13: Extract occlusion information by the state $\mathcal{S}$ of the object of k frames before occlusion;
14: Calculate the moving direction and distance of the object in the current frame;
15: Update weight $\{w_4, w_5, w_6, w_7\}$ of the full connection layer;
16: Obtain bounding box b_t of the object;
17: **end if**
18: **end for**

the center coordinate needs to be worked out

$$\bar{x}^{(t)} = x^{(t)} + \frac{1}{2}w^{(t)} \quad (7)$$

$$\bar{y}^{(t)} = y^{(t)} + \frac{1}{2}h^{(t)} \quad (8)$$

$$\bar{x}^{(t+1)} = x^{(t+1)} + \frac{1}{2}w^{(t+1)} \quad (9)$$

$$\bar{y}^{(t+1)} = y^{(t+1)} + \frac{1}{2}h^{(t+1)}. \quad (10)$$

The center distance between the two coordinates is

$$d^t = \sqrt{(\bar{x}^{(t+1)} - \bar{x}^{(t)})^2 + (\bar{y}^{(t+1)} - \bar{y}^{(t)})^2} \quad (11)$$

and the angle between the two coordinates is

$$\text{an}^t = \arctan \frac{\bar{y}^{(t+1)} - \bar{y}^{(t)}}{\bar{x}^{(t+1)} - \bar{x}^{(t)}}. \quad (12)$$

In addition, the distance positions $\{d^t, d^{t+1}, \dots, d^{t+n}\}$ and angles $\{\text{an}^t, \text{an}^{t+1}, \dots, \text{an}^{t+n}\}$ are saved to provide information for occlusion. This tightly links the context information to facilitate proper tracking.

2) *Object Appearance Model*: The appearance model of the object is learned by the network even during complete occlusion. k ($k = 50$ in the experiment) samples are randomly

selected as positive samples around the object bounding box, and $\text{IoU} > 0.7$. At the same time, k samples are randomly selected as negative samples in the region of $\text{IoU} < 0.3$. Compared with the previous frame, the positive sample and the negative sample remain the same under occlusion.

3) *Motion Vector*: The previous action dynamics vector needs to be saved for object tracking under complete occlusion. By storing the previous motion vectors, the tracker can make the correct action in the subsequent frame rather than blindly searching around the object bounding box under occlusion. According to the original motion trajectory of the object, including the movement direction and speed, the previous actions are saved to make tracking proceed in the original direction during complete occlusion. Using context information, the direction of motion is obtained for the current frame.

The distance positions $\{d^t, d^{t+1}, \dots, d^{t+n}\}$ and angles $\{\text{an}^t, \text{an}^{t+1}, \dots, \text{an}^{t+n}\}$ are obtained by the temporal and spatial information. Using the information in remote sensing video data, four pieces of occlusion information are acquired: the average transfer direction

$$\bar{\text{an}}^t = \frac{1}{n} \sum_{i=t}^{t+n} \text{an}^i \quad (13)$$

velocity

$$v = \frac{1}{n} \sum_{i=t}^{t+n} d^i \quad (14)$$

directional rotation change

$$c = \begin{cases} \frac{2}{n} \sum_{i=t}^{t+n} \frac{2}{n} (\text{an}^{i+\frac{n}{2}} - \text{an}^i), & \text{if } n \text{ is even} \\ \frac{2}{n-1} \sum_{i=t}^{t+n-1} \frac{2}{n-1} (\text{an}^{i+\frac{n-1}{2}} - \text{an}^i), & \text{if } n \text{ is odd} \end{cases} \quad (15)$$

and acceleration

$$a = \begin{cases} \frac{2}{n} \sum_{i=t}^{t+n} \frac{2}{n} (d^{i+\frac{n}{2}} - d^i), & \text{if } n \text{ is even} \\ \frac{2}{n-1} \sum_{i=t}^{t+n-1} \frac{2}{n-1} (d^{i+\frac{n-1}{2}} - d^i), & \text{if } n \text{ is odd.} \end{cases} \quad (16)$$

Through these calculations, the tracker can adapt to more datasets and thus has increased robustness. The robustness is evaluated by disrupting the initialization temporally (starting from a different frame) and spatially (start by a different bounding box) [13]. The direction and distance of the object bounding box to be moved can be obtained from this occlusion information in the current frame.

IV. EXPERIMENTS

In this section, we validate the performance of the proposed method through a series of experiments for object tracking on remote sensing datasets. All experiments are performed in Python3.5 using TensorFlow1.4 on a GeForce GTX 1080Ti with 32 GB of internal memory.

A. Implementation Details

Small CNN models are more effective than deep models in an object tracking task. Therefore, a pre-trained VGG-M model is chosen to initialize the proposed AD-OHNet. The DRL network includes three convolutional layers {conv1, conv2, conv3} and four fully connected layers {fc4, fc5, fc6, fc7}. During implementation, the input object image patch p of the DRL network has a dimension of $112 \times 112 \times 3$. The fully connected layers fc4 and fc5 have 512 output nodes. Finally, the fully connected layers fc6 and fc7 output the 11-D action probability and the 2-D confidence value, respectively.

The learning rate η_l is set to 0.003 in the experiments. During network fine-tuning, the number of iterations T_l is set to 300 at the first frame, and T_o is set to 30 for the online fine-tuning. Positive samples $N_{lp} = 180$ and negative samples $N_{ln} = 120$ are generated with 0.7 and 0.3 IoU overlap ratios in the first frame, respectively; meanwhile, positive samples $N_{op} = 32$ and negative samples $N_{on} = 18$ are generated with the same IoU overlap ratios in the other frame. In addition, the detection threshold T is set to 0.5. When the confidence value output by the fully connected layer is greater than the threshold, it is determined that the object has been correctly detected; otherwise, the tracking is considered to have failed.

B. Evaluation on Object Tracking Datasets

1) *Datasets*: Due to the scarcity and lack of remote sensing video datasets especially for complete occlusion, public datasets are utilized in the training process, and we fine-tune the network in the testing process. During the pretraining of the network, we use 360 training videos taken from some popular databases, i.e., VOT2013 [47], VOT2014 [48], VOT2015 [49], and ALOV300++ [50].

Remote sensing videos are used to test the performance of tracking, during which the network is fine-tuned to accommodate new data and changes. The remote sensing video sequences of Bogota, Hong Kong, and San Diego from Jilin-1, China's first commercial remote sensing satellites, are adopted for the test process. The resolutions of the Bogota, Hong Kong, and San Diego images are 0.92, 0.98, and 1.16 m, respectively. There are different object types in these video sequences, including airplane, car, ship, airport, port, and warship. The actual ground dimensions of the original videos of Bogota, Hong Kong, and San Diego are 55.2, 58.8, and 69.6 km², respectively. The duration of the three videos with 10 frames per second (FPS) is 33 s, and the image size of each frame is 12000×5000 pixels. The original image size is too large to extract features and, therefore, the region of interest in the videos is found, and the ground truth labels are annotated manually to achieve object tracking. In addition, there are some challenging situations, such as objects with straight and curved driving, small objects, complete occlusion during turning, two complete occlusions at different times in the same video, and the object being occluded by clouds. There are 100~330 frames in total, among which the number of occluded frames ranges from 10 to 33. Experiments on remote sensing videos of Bogota, Hong Kong, and San Diego validate

TABLE I

COMPARISON OF THE AD-OHNET TRACKER (BOTH WITH AND WITHOUT FINE-TUNING FOLLOWING TRACKING FAILURE) WITH THE BASELINE ADNET. THE TWO BEST RESULTS ARE SHOWN IN RED AND BLUE TEXT, RESPECTIVELY. THE AD-OHNET APPROACH SIGNIFICANTLY OUTPERFORMS ADNET, AND FINE-TUNING CAN IMPROVE TRACKING MEAN PRECISION (20 PIXELS) AND MEAN AUC

	Mean Precision	Mean AUC	Mean FPS
ADNet	0.658	0.632	3.272
Ours (no fine-tuning)	0.853	0.775	5.493
Ours (with fine-tuning)	0.96	0.842	2.971

the effectiveness and efficiency of the proposed AD-OHNet tracker.

2) *Evaluation Metrics*: Tracking performance is measured by three metrics, which are mean overlap precision, area under the curve (AUC), and tracking speed in FPS [7]. The mean overlap precision refers to the percentage of frames, in which the distance is higher than a certain threshold (usually 20 pixels [7], [51]), to all frames. The distance is the Euclidian distance between the ground truth and the predicted object center position of each frame. Meanwhile, the AUC is defined as the area under the curve in the success plot, which represents the IoU ratio between the ground truth and the bounding box obtained by the tracking algorithm.

C. Comparison to ADNet Baseline

In order to analyze the ability of the AD-OHNet tracker to solve complete occlusion, the proposed algorithm is compared with ADNet [42]. The results are reported in Table I.

In the experiment, occlusion handling is added based on the ADNet network. The confidence value of the network output is compared with the threshold T . If the confidence value is greater than the threshold T , it is determined that the object has been correctly detected in the current frame. If the confidence value is less than the threshold T , the object is considered to be in an occlusion state. The temporal and spatial contexts, object appearance model, and motion vector are adopted to drive actions of reinforcement learning under occlusion. It can be seen that the proposed tracker significantly outperforms ADNet for remote sensing videos under occlusion (see Table I).

We also compare the effect of fine-tuning during occlusion handling. By fixing the weight parameters of the convolutional layer {conv1, conv2, conv3} of the network and adjusting the weight parameters of all fully connected layers {fc4, fc5, fc6, fc7}, the network is fine-tuned online to enhance the robustness of tracking. The online update adaptation proposed by Nam and Han [8] is adopted. The AD-OHNet is fine-tuned every m -frames or when the tracking fails. The full connection layers of the network are updated by extracting features from positive samples and negative samples. The convolutional layers have generic tracking information, whereas the fully connected layers contain video-specific knowledge. The learning rate η_o is set to 0.001 for the online fine-tuning. It can be seen from Table I that although the online fine-tuning of the network reduces the detection speed,

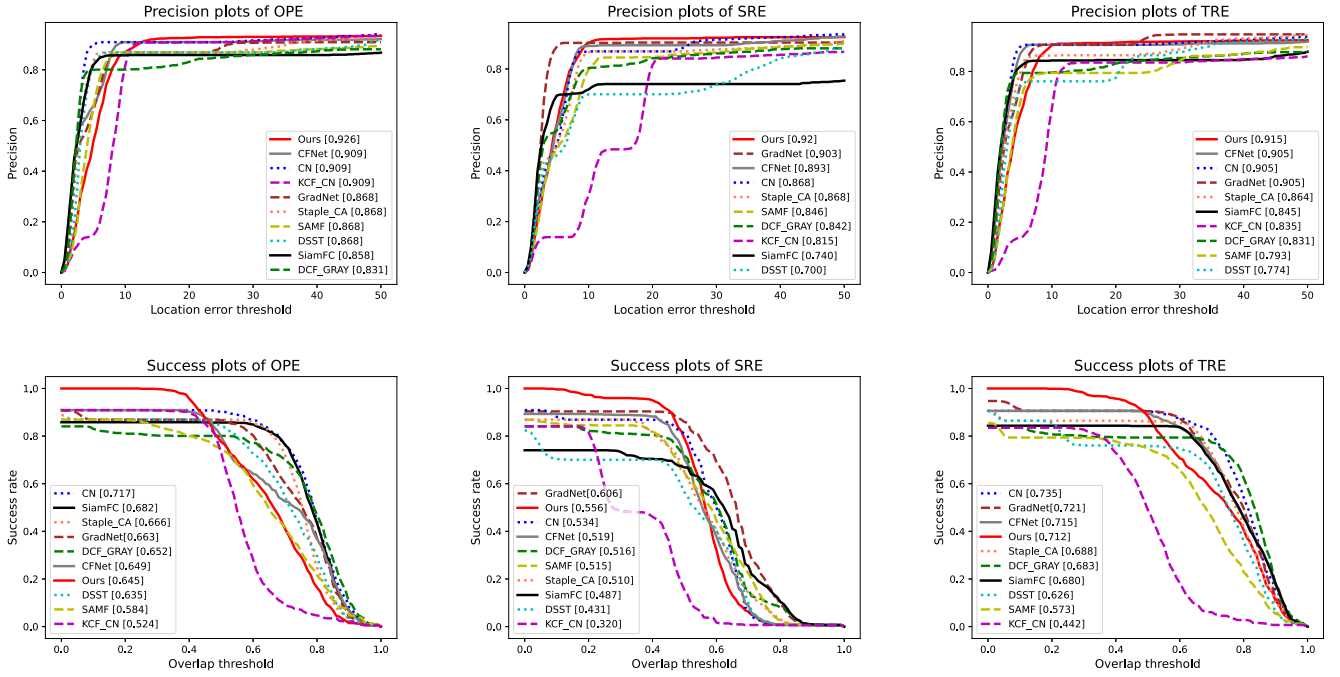


Fig. 7. Precision plots and success plots for OPE, SRE, and TRE on the remote sensing video datasets with completely occluded objects. We compare the top 10 trackers of 22 trackers. The values in the legend indicate the precision at a threshold of 20 pixels for the precision plot and the AUC score for the success plot. Our approach demonstrates superior robustness compared to other methods.

the detection precision increases. In summary, the proposed tracker improves the tracking precision by 30.2 percentage points (p.p.) while reducing the tracking speed by 0.301 FPS.

D. Comparison With State-of-the-Art Trackers

In this section, we compare the AD-OHNet approach with nine state-of-the-art trackers on remote sensing video benchmarks with completely occluded objects. The trackers used for comparison are as follows: CFNet [24], GradNet [27], CN [15], SAMF [52], SiamFC [23], DSST [7], Staple_CA [53], KCF_CN [54], and DCF_GRAY [6]. The precision plots and success plots of one-pass evaluation (OPE) are shown in Fig. 7, which include the top 10 trackers selected from 22 trackers. The OPE initializes the first frame with the position of the object in the ground truth and then runs a tracker throughout a test sequence to obtain the mean precision and success rate. The mean precision scores and AUC of the trackers on the sequences are reported in the figure legends.

The comparison shows that the proposed AD-OHNet achieves superior performance compared with the existing trackers. These outstanding results are partly attributed to the strength of occlusion handling. Most trackers will not be able to accurately locate the object after encountering a complete occlusion video sequence. Because the object appears occluded in the frame, it will not be tracked in the subsequent frames. Therefore, some identical precision values often appear in Fig. 7. However, precision can reach 92.6% when the threshold is 20 pixels in the AD-OHNet tracker. Although the AUC score is not the best in the success plot, the proposed method has the highest success rate when the overlap threshold is lower than 0.4. It can be seen from the precision plots and success plots that the AD-OHNet approach

can effectively solve the problem of occlusion in remote sensing video sequences.

Traditionally, a more complex network model or strategy is usually adopted to achieve redetection of the object, which increases computational complexity for object tracking under occlusion. For example, Li *et al.* [34] proposed a long-term tracking strategy to deal with occlusion. Their tracking system is composed of two parts: the context-aware correlation filter tracker, which is an efficient correlation method for short-term tracking, and the SVM-based redetector, which prevents the context-aware tracker from degradation. However, we propose an end-to-end tracking algorithm without an additional network structure. The temporal and spatial contexts between consecutive frames in a remote sensing sequence, the object appearance model learned by the network, and the motion vector obtained from action parameters in reinforcement learning are adopted to provide the occlusion information. When the network is fine-tuned online, the parameters of the convolutional layer $\{w_1, w_2, w_3\}$ are fixed, and the parameters of the fully connected layer $\{w_4, w_5, w_6, w_7\}$ are fine-tuned to adapt to the shape change and deformation of the object in the test sequences; this makes our tracking algorithm more robust. It can be seen from Table I that our tracker improves the tracking precision by 30.2 p.p. while reducing the tracking speed by only 0.301 FPS due to the online fine-tuning of the network. Therefore, the proposed AD-OHNet tracker achieves low computational complexity for object tracking under occlusion.

Fig. 8 shows the qualitative results comparing the AD-OHNet with the five other trackers on several sequences of remote sensing videos with completely occluded objects. The videos show some challenging situations, such as objects with

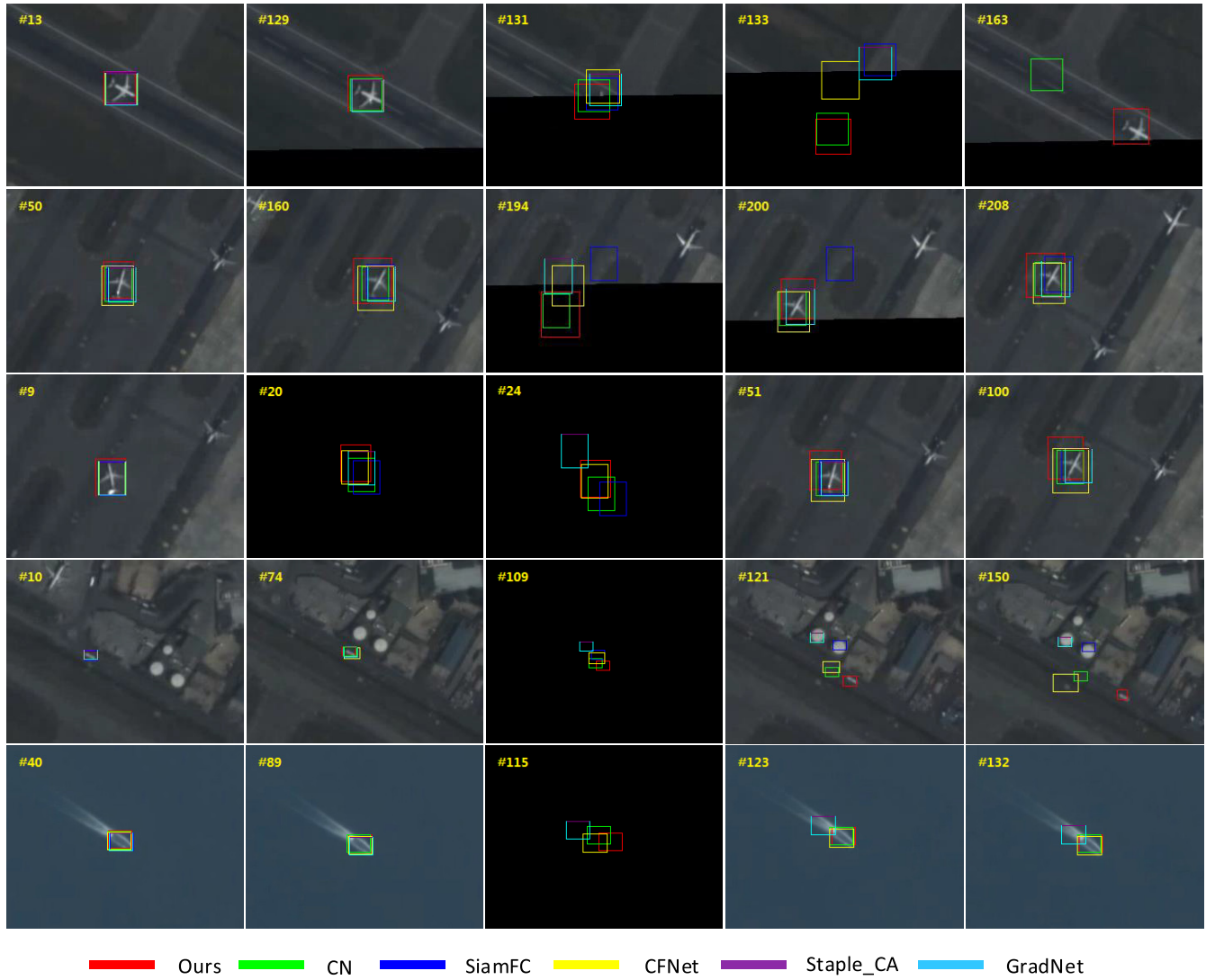


Fig. 8. Qualitative results comparing the AD-OHNet approach with other trackers on several sequences of a remote sensing video dataset with completely occluded objects. The number in the top-left corner indicates the frame number within the video. The results show some challenging situations, which include an airplane flying straight, an airplane with curved flying, an airplane with complete occlusion while turning, a small moving car, and a moving ship. The proposed tracker performs favorably compared with the existing trackers under these tricky situations.

straight and curved driving, small objects, and even complete occlusion during turning.

The proposed tracker performs favorably compared with the existing trackers under some tricky situations. The first line of images in Fig. 8 shows the tracking results of occlusion handling for the airplane flying in a straight line. The second line of images shows the tracking results of occlusion handling for the airplane with curved flying. The third line of images shows the tracking results of occlusion handling when the airplane is turning. The fourth line of images shows the tracking results of occlusion handling for the small moving car. The remote sensing image is 285×220 pixels, but the object's size is only 28×10 pixels. The fifth line of images shows the tracking results of occlusion handling for the moving ship. The proposed AD-OHNet tracker performs excellently compared with the other trackers under these tricky situations.

E. Comparison of Robustness to Initialization

We evaluate the robustness of the AD-OHNet with respect to initialization. In addition to the OPE, there are two different initialization evaluations for robustness measurement: spatial robustness evaluation (SRE) and temporal robustness evaluation (TRE). The SRE extracts the initial bounding box by moving or scaling the accurate ground truth in the first frame. The TRE means that the object starts to be tracked from different moments in a video sequence. The precision plots and success plots are shown for the TRE and SRE in Fig. 7.

As can be seen from Fig. 7, the proposed tracker can effectively improve tracking performance compared to other methods in both initialization evaluations. Precision and AUC are presented for the AD-OHNet tracker in the precision plots and success plots, respectively. Due to the occlusion handling program, the AD-OHNet can accurately predict the position

TABLE II

COMPARISON OF OUR TRACKER WITH STATE-OF-THE-ART TRACKERS ON REMOTE SENSING VIDEOS WITH COMPLETELY OCCLUDED OBJECTS. THE TWO BEST RESULTS ARE SHOWN IN RED AND BLUE TEXT, RESPECTIVELY. OUR AD-OHNET APPROACH ACHIEVES SUPERIOR PERFORMANCE COMPARED WITH OTHER TRACKERS

Trackers	OPE		SRE		TRE	
	Precision	AUC	Precision	AUC	Precision	AUC
CFNet [24]	0.909	0.649	0.893	0.519	0.905	0.715
GradNet [27]	0.868	0.663	0.903	0.606	0.905	0.721
CN [15]	0.909	0.717	0.868	0.534	0.905	0.735
SAMF [52]	0.868	0.584	0.846	0.515	0.793	0.573
SiamFC [23]	0.858	0.682	0.740	0.487	0.845	0.680
SiamBAN [28]	0.818	0.616	0.784	0.552	0.892	0.668
DSST [7]	0.868	0.635	0.700	0.431	0.774	0.626
Staple_CA [53]	0.868	0.666	0.868	0.510	0.864	0.688
KCF_CN [54]	0.909	0.524	0.815	0.320	0.835	0.442
DCF_GRAY [6]	0.831	0.652	0.842	0.516	0.831	0.683
BACF [55]	0.700	0.546	0.700	0.435	0.690	0.554
CSK [5]	0.800	0.593	0.701	0.418	0.796	0.569
CSR-DCF [56]	0.700	0.525	0.700	0.442	0.690	0.552
DAT [57]	0.700	0.467	0.656	0.375	0.665	0.382
DCF_HOG [6]	0.740	0.485	0.778	0.480	0.731	0.538
KCF_GRAY [6]	0.831	0.650	0.842	0.516	0.836	0.666
KCF_HOG [6]	0.840	0.532	0.758	0.456	0.799	0.577
MCCT-H [58]	0.700	0.556	0.700	0.408	0.690	0.562
MOSSE [4]	0.696	0.461	0.700	0.409	0.690	0.439
Staple [21]	0.868	0.671	0.800	0.490	0.864	0.689
STRCF [59]	0.700	0.553	0.700	0.447	0.690	0.567
Ours	0.926	0.645	0.920	0.556	0.915	0.712

and direction of the object's motion even if the object is in the state of complete occlusion. Although some bounding boxes are not very accurate, the proposed tracker can still locate the object correctly, while other trackers fail. The results demonstrate the superior robustness of the proposed approach.

In addition, Table II provides a comparison of precision and AUC over all videos on the remote sensing dataset with completely occluded objects. It is worth mentioning that the precisions in Table II are calculated for all frames other than occluded frames. The tracking evaluation metrics of 22 tracking methods under three different initialization conditions on remote sensing video sequences with complete occlusion are shown. As shown in Table II, the proposed approach achieves superior performance compared with other trackers.

However, the proposed tracking method still cannot solve some complex situations. Fig. 9 shows a failure case for a ship occluded by three clouds. When occluded by the first and second clouds, the proposed approach can still accurately track the object. However, the tracking fails after the object is occluded by the third cloud. There is not only a long-time occlusion (about 80 frames) but also a change in the direction and speed of the object during the occlusion. Both the satellite

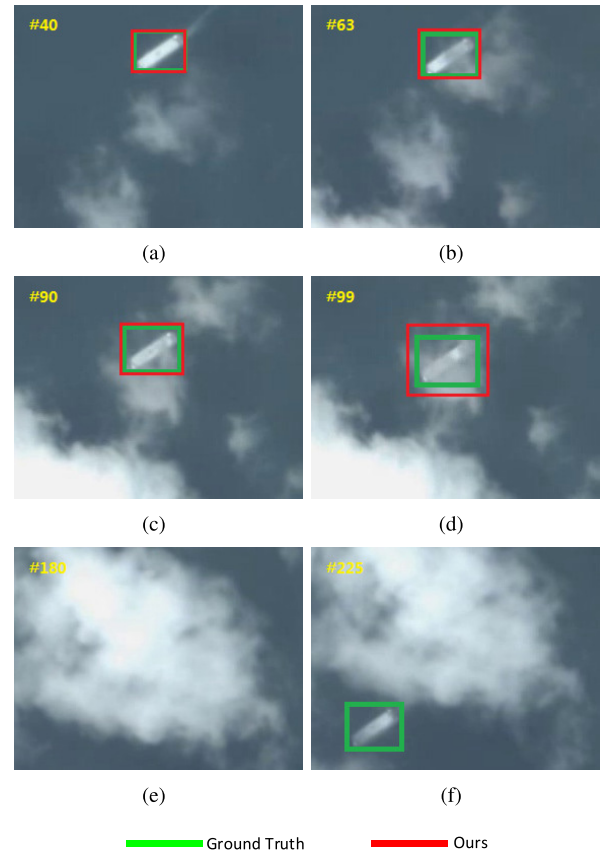


Fig. 9. Failure case of the proposed AD-OHNet method on a remote sensing video with a ship completely occluded by clouds. (a) Tracking results at frame 40. (b) Tracking results of the ship partially occluded by the first cloud. (c) Tracking results after the ship is partially occluded by the first cloud. (d) Tracking results of the ship occluded by the second cloud. (e) Tracking results of the ship completely occluded by the third cloud. (f) Failure tracking results after the ship is completely occluded by the third cloud.

and object are moving at the same time in the video, which greatly increases the difficulty of object tracking.

V. CONCLUSION

In this article, we propose an object tracking method based on DRL to solve the problem of object complete occlusion in remote sensing videos. The proposed algorithm achieves good tracking performance, even for some tricky situations on a remote sensing dataset. The temporal and spatial contexts, object appearance model, and motion vector are adopted to drive actions of reinforcement learning under occlusion. The effectiveness of the proposed tracker is demonstrated by the experimental results.

However, there are several aspects to be improved. First of all, there are generally multiple objects in a remote sensing video. In the future, multiple-object tracking methods based on DRL can be proposed to deal with the occlusion problem in remote sensing videos. Second, this study still cannot solve the problem of object occlusion in certain complex scenes. For example, objects are occluded caused by the movement of the tracking lens or clouds. Both the satellite and object are moving at the same time under these circumstances. In such a situation, the object's movement direction and speed will change during occlusion. In the future, more complex network

models can be utilized. Finally, the objects tracked in the pretrained network may be inconsistent with the actual tracked objects. More complete pretraining models can be built for different datasets in the future.

REFERENCES

- [1] B. Ristic, S. Arulampalam, and N. Gordon, "Beyond the Kalman filter," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 19, no. 7, pp. 37–38, Jul. 2004.
- [2] F. Gustafsson *et al.*, "Particle filters for positioning, navigation, and tracking," *IEEE Trans. Signal Process.*, vol. 50, no. 2, pp. 425–437, Feb. 2002.
- [3] D. Comaniciu, V. Ramesh, and P. Meer, "Real-time tracking of non-rigid objects using mean shift," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 2, Jun. 2000, pp. 142–149.
- [4] D. Bolme, J. R. Beveridge, B. A. Draper, and Y. M. Lui, "Visual object tracking using adaptive correlation filters," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2010, pp. 2544–2550.
- [5] J. F. Henriques, R. Caseiro, P. Martins, and J. Batista, "Exploiting the circulant structure of tracking-by-detection with kernels," in *Proc. Eur. Conf. Comput. Vis. Berlin, Germany: Springer*, 2012, pp. 702–715.
- [6] J. F. Henriques, R. Caseiro, P. Martins, and J. Batista, "High-speed tracking with kernelized correlation filters," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 3, pp. 583–596, Mar. 2015.
- [7] M. Danelljan, G. Häger, F. S. Khan, and M. Felsberg, "Discriminative scale space tracking," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 8, pp. 1561–1575, Aug. 2017.
- [8] H. Nam and B. Han, "Learning multi-domain convolutional neural networks for visual tracking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 4293–4302.
- [9] M. Danelljan, A. Robinson, F. S. Khan, and M. Felsberg, "Beyond correlation filters: Learning continuous convolution operators for visual tracking," in *Proc. Eur. Conf. Comput. Vis. Cham, Switzerland: Springer*, 2016, pp. 472–488.
- [10] M. Danelljan, G. Bhat, F. S. Khan, and M. Felsberg, "ECO: Efficient convolution operators for tracking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 6638–6646.
- [11] A. Yilmaz, X. Li, and M. Shan, "Contour-based object tracking with occlusion handling in video acquired using mobile cameras," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 26, no. 11, pp. 1531–1536, Nov. 2004.
- [12] L. A. Camuñas-Mesa, T. Serrano-Gotarredona, S.-H. Ieng, R. Benosman, and B. Linares-Barranco, "Event-driven stereo visual tracking algorithm to solve object occlusion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 9, pp. 4223–4237, Sep. 2018.
- [13] Y. Wu, J. Lim, and M.-H. Yang, "Online object tracking: A benchmark," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2013, pp. 2411–2418.
- [14] M. Fiaz, A. Mahmood, S. Javed, and S. K. Jung, "Handcrafted and deep trackers: Recent visual object tracking approaches and trends," *ACM Comput. Surv.*, vol. 52, no. 2, pp. 1–44, May 2019.
- [15] M. Danelljan, F. S. Khan, M. Felsberg, and J. Van De Weijer, "Adaptive color attributes for real-time visual tracking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 1090–1097.
- [16] M. Danelljan, G. Häger, F. S. Khan, and M. Felsberg, "Accurate scale estimation for robust visual tracking," in *Proc. Brit. Mach. Vis. Conf.*, 2014, pp. 1–11.
- [17] M. Danelljan, G. Hager, F. S. Khan, and M. Felsberg, "Learning spatially regularized correlation filters for visual tracking," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2015, pp. 4310–4318.
- [18] T. Zhang, C. Xu, and M.-H. Yang, "Learning multi-task correlation particle filters for visual tracking," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 2, pp. 365–378, Feb. 2019.
- [19] S. Hong, T. You, S. Kwak, and B. Han, "Online tracking by learning discriminative saliency map with convolutional neural network," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 597–606.
- [20] H. Nam, M. Baek, and B. Han, "Modeling and propagating CNNs in a tree structure for visual tracking," 2016, *arXiv:1608.07242*. [Online]. Available: <http://arxiv.org/abs/1608.07242>
- [21] L. Bertinetto, J. Valmadre, S. Golodetz, O. Miksik, and P. H. S. Torr, "Staple: Complementary learners for real-time tracking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 1401–1409.
- [22] Y. Lu, Y. Yuan, and Q. Wang, "A dense connection based network for real-time object tracking," *Neurocomputing*, vol. 410, pp. 229–236, Oct. 2020.
- [23] L. Bertinetto, J. Valmadre, J. F. Henriques, A. Vedaldi, and P. H. Torr, "Fully-convolutional Siamese networks for object tracking," in *Proc. Eur. Conf. Comput. Vis. Cham, Switzerland: Springer*, 2016, pp. 850–865.
- [24] J. Valmadre, L. Bertinetto, J. Henriques, A. Vedaldi, and P. H. S. Torr, "End-to-end representation learning for correlation filter based tracking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2805–2813.
- [25] Y. Zhang, L. Wang, J. Qi, D. Wang, M. Feng, and H. Lu, "Structured Siamese network for real-time visual tracking," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 351–366.
- [26] Q. Wang, Z. Teng, J. Xing, J. Gao, W. Hu, and S. Maybank, "Learning attentions: Residual attentional Siamese network for high performance online visual tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4854–4863.
- [27] P. Li, B. Chen, W. Ouyang, D. Wang, X. Yang, and H. Lu, "GradNet: Gradient-guided network for visual object tracking," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6162–6171.
- [28] Z. Chen, B. Zhong, G. Li, S. Zhang, and R. Ji, "Siamese box adaptive network for visual tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 6668–6677.
- [29] Z. Yin and R. T. Collins, "Object tracking and detection after occlusion via numerical hybrid local and global mode-seeking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2008, pp. 1–8.
- [30] Z. H. Khan and I. Y.-H. Gu, "Nonlinear dynamic model for visual object tracking on Grassmann manifolds with partial occlusion handling," *IEEE Trans. Cybern.*, vol. 43, no. 6, pp. 2005–2019, Dec. 2013.
- [31] S. Zhao, S. Zhang, and L. Zhang, "Towards occlusion handling: Object tracking with background estimation," *IEEE Trans. Cybern.*, vol. 48, no. 7, pp. 2086–2100, Jul. 2018.
- [32] P. Zhang *et al.*, "Robust visual tracking using multi-frame multi-feature joint modeling," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 29, no. 12, pp. 3673–3686, Dec. 2019.
- [33] M. Guan, C. Wen, M. Shan, C.-L. Ng, and Y. Zou, "Real-time event-triggered object tracking in the presence of model drift and occlusion," *IEEE Trans. Ind. Electron.*, vol. 66, no. 3, pp. 2054–2065, Mar. 2019.
- [34] T. Li, S. Zhao, Q. Meng, Y. Chen, and J. Shen, "A stable long-term object tracking method with re-detection strategy," *Pattern Recognit. Lett.*, vol. 127, pp. 119–127, Nov. 2019.
- [35] Q. Zhou, B. Zhong, Y. Zhang, J. Li, and Y. Fu, "Deep alignment network based multi-person tracking with occlusion and motion reasoning," *IEEE Trans. Multimedia*, vol. 21, no. 5, pp. 1183–1194, May 2018.
- [36] F. Wu, J. Zhang, and Z. Xu, "Stably adaptive anti-occlusion Siamese region proposal network for real-time object tracking," *IEEE Access*, vol. 8, pp. 161 349–161 360, 2020.
- [37] Y. Yuan, J. Chu, L. Leng, J. Miao, and B.-G. Kim, "A scale-adaptive object-tracking algorithm with occlusion detection," *EURASIP J. Image Video Process.*, vol. 2020, no. 1, pp. 1–15, Dec. 2020.
- [38] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*. Cambridge, MA, USA: MIT Press, 2018.
- [39] K. Arulkumaran, M. P. Deisenroth, M. Brundage, and A. A. Bharath, "Deep reinforcement learning: A brief survey," *IEEE Signal Process. Mag.*, vol. 34, no. 6, pp. 26–38, Nov. 2017.
- [40] J. Supancic, III, and D. Ramanan, "Tracking as online decision-making: Learning a policy from streaming videos with reinforcement learning," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 322–331.
- [41] Y. Xie, J. Xiao, K. Huang, J. Thiyagalingam, and Y. Zhao, "Correlation filter selection for visual tracking using reinforcement learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 1, pp. 192–204, Jan. 2020.
- [42] S. Yun, J. Choi, Y. Yoo, K. Yun, and J. Y. Choi, "Action-driven visual object tracking with deep reinforcement learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 6, pp. 2239–2252, Jun. 2018.
- [43] X. Dong, J. Shen, W. Wang, Y. Liu, L. Shao, and F. Porikli, "Hyperparameter optimization for tracking with continuous deep Q-learning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 518–527.
- [44] W. Luo, P. Sun, F. Zhong, W. Liu, T. Zhang, and Y. Wang, "End-to-end active object tracking and its real-world deployment via reinforcement learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 6, pp. 1317–1332, Jun. 2020.
- [45] B. Fukun, L. Mingyang, and S. Jiayi, "A robust and effective tracking method in remote sensing video sequences," in *Proc. 2nd Int. Conf. Control Comput. Vis.*, 2019, pp. 13–17.

- [46] Z. Teng, J. Xing, Q. Wang, C. Lang, S. Feng, and Y. Jin, "Robust object tracking based on temporal and spatial deep networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 1144–1153.
- [47] M. Kristan *et al.*, "The visual object tracking VOT2013 challenge results," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops*, Dec. 2013, pp. 98–111.
- [48] M. Kristan *et al.*, "The visual object tracking VOT2014 challenge results," in *Proc. Eur. Conf. Comput. Vis.*, vol. 8926, 2014, pp. 191–217.
- [49] M. Kristan *et al.*, "The visual object tracking VOT2015 challenge results," in *Proc. IEEE Int. Conf. Comput. Vis. Workshop (ICCVW)*, Dec. 2015, pp. 564–586.
- [50] A. W. M. Smeulders, D. M. Chu, R. Cucchiara, S. Calderara, A. Dehghan, and M. Shah, "Visual tracking: An experimental survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 36, no. 7, pp. 1442–1468, Jul. 2014.
- [51] B. Babenko, M.-H. Yang, and S. Belongie, "Robust object tracking with online multiple instance learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 8, pp. 1619–1632, Aug. 2011.
- [52] Y. Li and J. Zhu, "A scale adaptive kernel correlation filter tracker with feature integration," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 254–265.
- [53] M. Mueller, N. Smith, and B. Ghanem, "Context-aware correlation filter tracking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 1396–1404.
- [54] C. S. Asha and A. V. Narasimhadhan, "Adaptive learning rate for visual tracking using correlation filters," *Procedia Comput. Sci.*, vol. 89, pp. 614–622, Jan. 2016.
- [55] H. K. Galoogahi, A. Fagg, and S. Lucey, "Learning background-aware correlation filters for visual tracking," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 1135–1143.
- [56] A. Lukežič, T. Vojř, L. Ehovinzajc, J. Matas, and M. Kristan, "Discriminative correlation filter tracker with channel and spatial reliability," *Int. J. Comput. Vis.*, vol. 126, no. 7, pp. 671–688, Jul. 2018.
- [57] H. Possegger, T. Mauthner, and H. Bischof, "In defense of color-based model-free tracking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 2113–2120.
- [58] N. Wang, W. Zhou, Q. Tian, R. Hong, M. Wang, and H. Li, "Multi-cue correlation filters for robust visual tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 4844–4853.
- [59] F. Li, C. Tian, W. Zuo, L. Zhang, and M.-H. Yang, "Learning spatial-temporal regularized correlation filters for visual tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 4904–4913.



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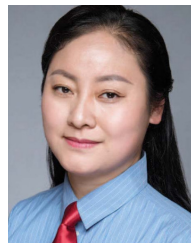
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